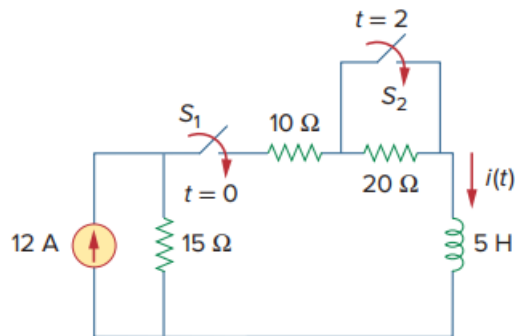


Problem

Switch S_1 in Fig. 7.54 is closed at $t = 0$, and switch S_2 is closed at $t = 2$ s. Calculate $i(t)$ for all t . Find $i(1)$ and $i(3)$.

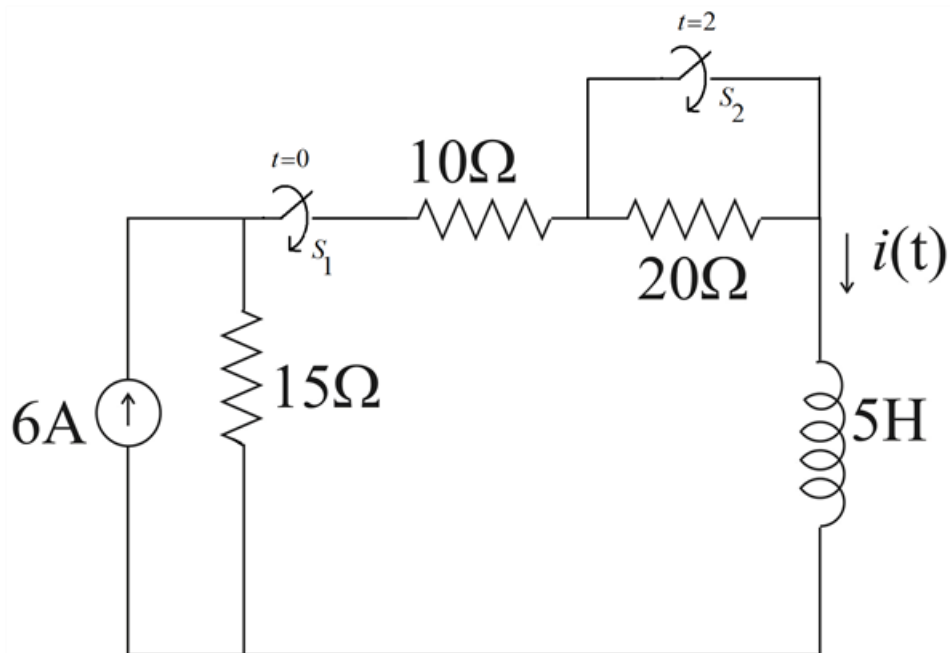
Fig. 7.54:



Step-by-step solution

Step 1 of 11

Given circuit diagram is



Step 2 of 11

We need to consider three intervals $t \leq 0$, $0 \leq t \leq 2$ and $t > 2$ sec

For $t < 0$ switches S_1 and S_2 are open so that $i = 0$

Since the inductor current can't change instantaneously

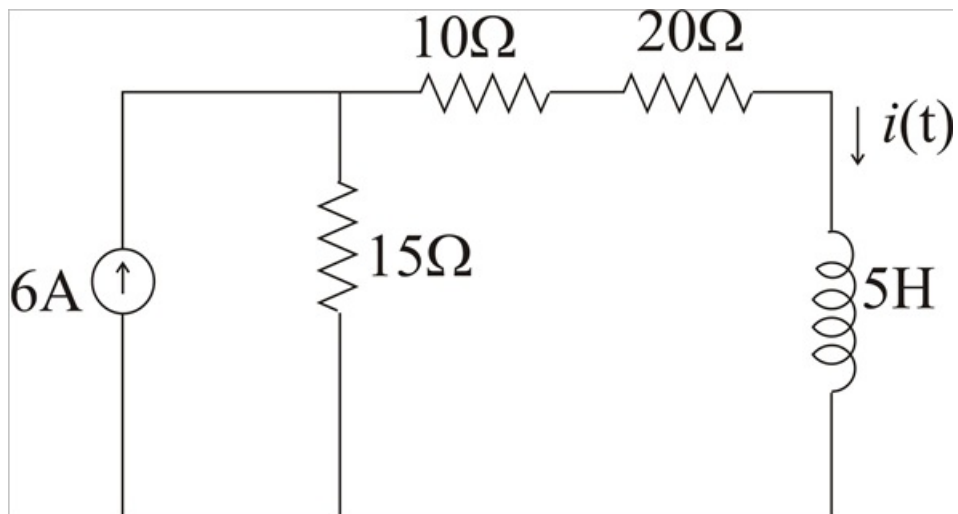
$$i(0^-) = i(0^+) = i(0) = 0 \text{ A}$$

Step 3 of 11

For $0 \leq t \leq 2 \text{ sec}$ S_1 is closed and S_2 opened

For $0 \leq t \leq 2 \text{ sec}$ $0 \leq t \leq 2 \text{ sec}$ given circuit is as below

Step 4 of 11



Step 5 of 11

Assume for now that S_1 is closed forever, inductor becomes short circuited

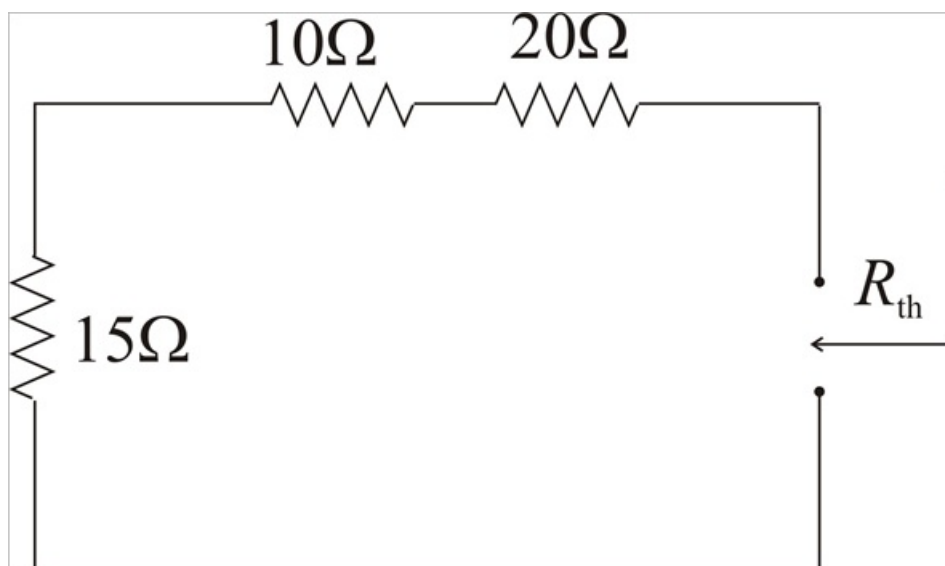
$i(\infty)$ can be found by using current division rule

$$\therefore i(\infty) = 6 \times \left(\frac{15}{15 + 10 + 20} \right)$$

$$i(\infty) = 2 \text{ A}$$

Equivalent or Thevenin's resistance R_{th} by looking onto inductor terminals can be calculated by open circuiting current source

Step 6 of 11



Step 7 of 11

$$R_{th} = 45\Omega$$

∴ Time constant

$$\tau = \frac{L}{R_{th}}$$

$$\tau = \frac{5}{45}$$

$$\tau = \frac{1}{9} \text{ sec}$$

∴ Inductor current

$$i(t) = i(\infty) + [i(0) - i(\infty)]e^{-\frac{t}{\tau}}$$

$$i(t) = 2 + (0 - 2)e^{-\left(\frac{1}{9}\right)t} \text{ A for } 0 < t \leq 2 \text{ sec}$$

$$i(t) = 2(1 - e^{-9t}) \text{ A for } 0 < t < 2 \text{ sec}$$

Step 8 of 11

For $t \geq 2 \text{ sec}$, switch S_2 is closed and as a result 20Ω Resistor is short circuited. This sudden change will not affect the inductor current because inductor current cannot change instantaneously

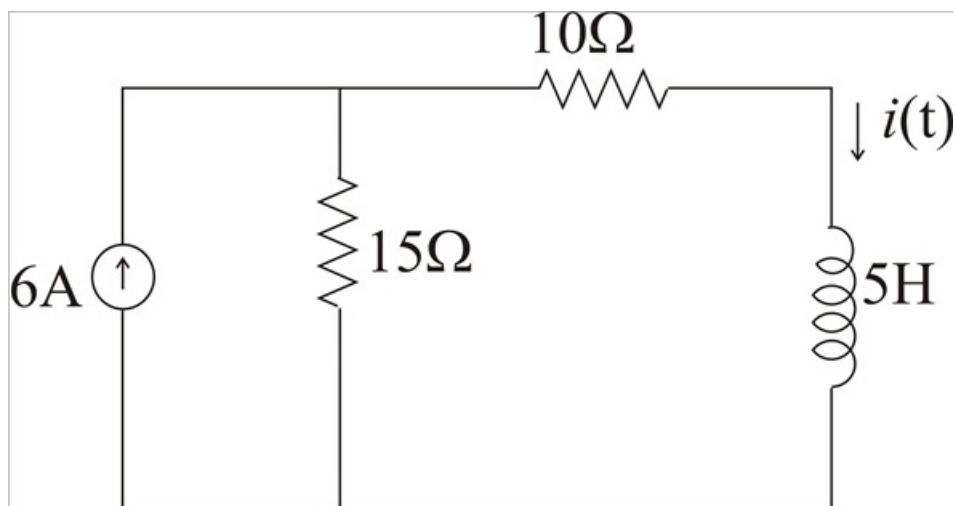
$$i(2) = i(2^-) = 2(1 - e^{-9 \times 2})$$

$$i(2) = 2(1 - 15.23 \times 10^{-9})$$

$$i(2) = 2 \text{ A}$$

Step 9 of 11

For $t \geq 2 \text{ sec}$ circuit is



Step 10 of 11

$i(\infty)$ Can be found by short circuiting the inductor

$$\therefore i(\infty) = 6 \times \left(\frac{15}{15+10} \right)$$

$$i(\infty) = 3.6 \text{ A}$$

Equivalent or Thevenin's resistance R_{th} by looking onto inductor terminals is

$$R_{th} = 15 + 20$$

$$R_{th} = 25 \Omega$$

Inductor current

$$i(t) = i(\infty) + [i(2) - i(\infty)] e^{-\frac{(t-2)}{\tau}}$$

Time constant

$$\tau = \frac{L}{R}$$

$$\tau = \frac{5}{25}$$

$$\tau = \frac{1}{5} \text{ sec}$$

Step 11 of 11

$$i(t) = 3.6 + (2 - 3.6) e^{-\frac{(t-2)}{5}} \text{ A for } t \geq 2 \text{ sec}$$

$$i(t) = 3.6 - 1.6 e^{-5(t-2)} \text{ A for } t \geq 2 \text{ sec}$$

$$\therefore i(t) = \begin{cases} 0 & t \leq 0 \\ 2(1 - e^{-9t}) \text{ A} & 0 \leq t \leq 2 \text{ sec} \\ 3.6 - 1.6 e^{-5(t-2)} \text{ A} & t \geq 2 \text{ sec} \end{cases}$$

$i(1)$ can be calculated from equation for $0 \leq t \leq 2 \text{ sec}$

$$\therefore i(1) = 2(1 - e^{-9})$$

$$\boxed{i(1) = 1.9997 \text{ A}}$$

$i(3)$ can be calculated from equation for $t \geq 2 \text{ sec}$

$$i(3) = 3.6 - 1.6 e^{-5(3-2)}$$

$$\boxed{i(3) = 3.589 \text{ A}}$$